

```

1  n = input("Enter the number of rows: ");
2  m = input("Enter the number of columns: ");
3
4  k = input("Enter the number of cells: ");
5
6  coords = zeros(k, 3);
7  gridd = zeros(n, m);
8
9  for i=1:k
10     coords(i,1:2) = input(sprintf("Enter cell %d: ", i));
11     coords(i,3) = 1;
12     gridd(coords(i,1), coords(i,2)) = 1;
13 end
14
15 gridd
16
17 for i=1:k
18     ai = coords(i,1);
19     aj = coords(i,2);
20
21     if coords(i,3) == -1
22         continue
23     end
24
25     % coords(i,3) = 2;
26
27     grid_cpy = gridd;
28     if ~find_loop_col(ai,aj,k, grid_cpy, coords, ai, aj)
29         coords(i,3) = -1;
30     else
31         disp([ai, aj])    %%%%%%%%%%%%%%%
32         return;
33     end
34 end
35
36 disp("No Loop was found!")
37
38 %%
39
40 function found_bool = find_loop_col(r,c,k, grid, coords, ai, aj)
41     % disp([r,c])
42     found_bool = false;
43     in_this_col = false;
44     for coord=1:k
45         if coords(coord, 2) == c && coords(coord, 3) == 1
46             coords(coord, 3) = 2;
47             in_this_col = true;
48             r = coords(coord, 1);
49             if find_loop_row(r,c,k, grid, coords, ai, aj)
50                 found_bool = true;    %%%%%%%%%%%%%%%
51                 disp([r,c])
52                 return
53             end
54             coords(coord, 3) = 1;
55         end
56     end
57     if ~in_this_col
58         return

```

```

59     end
60
61 end
62
63 function found_bool = find_loop_row(r,c,k, grid, coords, ai, aj)
64     % disp([r,c])
65     found_bool = false;
66     in_this_row = false;
67     for coord=1:k
68         if coords(coord, 1) == r && coords(coord, 3) == 1
69             coords(coord, 3) = 2;
70             in_this_row = true;
71             c = coords(coord, 2);
72             if r == ai && c == aj
73                 disp("Loop found!")
74                 found_bool = true;
75                 return
76             end
77             if find_loop_col(r,c,k, grid, coords, ai, aj)
78                 found_bool = true;
79                 disp([r,c]) %%%%%%%%%%%%%%%
80                 return
81             end
82             coords(coord, 3) = 1;
83         end
84     end
85     if ~in_this_row
86         return
87     end
88
89 end

```